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论文: multiagents.org/2026_papers/socially_aware_multi_agent.pdf

近两年代表论文

- [AAAI 2026] Toward Socially Aware Multi-Agent Systems: Measuring Group-Level Influence of LLM Agents
- [2025] Social Dynamics in LLM-Based Multi-Agent Systems
- [2025] Measuring Influence and Persuasion in Agent Teams
- [2024] Group Decision Making with Large Language Models
- [2024] Emergent Behaviors in Multi-Agent LLM Systems

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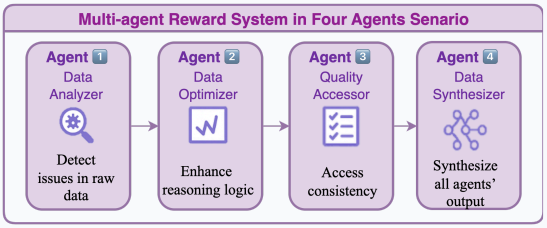
Toward Socially Aware Multi-Agent Systems: Measuring Group-Level Influence of LLM Agents

创新点总结

核心贡献：本文提出了一种测量LLM多智能体系统中群体层面影响力的框架，填补了多智能体协作研究中社会动力学分析的空白。创新包括：(1)定义了群体影响力指标，量化单个Agent对团队决策的影响程度；(2)设计了实验范式，系统评估不同Agent角色和位置对最终结果的影响；(3)揭示了LLM Agent在团队协作中的说服策略和影响力传播机制；(4)提出了改善多智能体系统公平性和多样性的建议。该工作为构建更加社会感知的多智能体系统奠定了理论基础，对理解LLM Agent的团队行为模式具有重要意义。

影响力	社会	群体	WMAC
量化框架	动力学	层面分析	Oral



群体影响力测量框架：多智能体系统中的社会动力学分析

合作建议

NUS在多智能体社会动力学方向有深厚积累，适合在Agent协作、群体决策、社会计算上合作。WMAC是AAAI 2026 Multi-Agent方向核心Workshop。

P

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Gradient · Columbia · Waseda

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论文: arxiv.org/abs/2511.16202

近两年代表论文

- [AAAI 2026] Multi-Agent Collaborative Reward Design for Enhancing Reasoning in Reinforcement Learning
- [2025] Collaborative Reward Modeling with Multi-Agent Systems
- [2025] RewardBench: A Benchmark for Collaborative Reward Design
- [2024] Enhancing RL Reasoning with Multi-Agent Collaboration
- [2024] Transparent Reward Modeling for Reinforcement Learning

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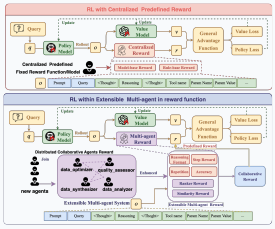
Multi-Agent Collaborative Reward Design for Enhancing Reasoning in Reinforcement Learning

创新点总结

核心贡献：本文提出了CRM（Collaborative Reward Design）框架，将多智能体协作引入强化学习的奖励设计过程。创新包括：(1)设计了协作式奖励建模架构，多个Agent分别负责不同维度的奖励信号生成；(2)引入了RewardBench基准，用于评估和训练协作奖励模型；(3)实现了更加透明和稳定的奖励优化过程，解决了传统RL奖励设计的稀疏和不稳定问题；(4)在复杂推理任务上展示了显著性能提升。该工作将多智能体思想应用于RL核心难题，为构建更可靠的RL系统提供了新思路。

协作	透明	Reward	多机构
奖励设计	优化	Bench	合作



CRM框架：多智能体协作式奖励设计方法

合作建议

Gradient、Columbia、Waseda联合出品，多机构资源丰富。适合在RL奖励设计、多智能体协作、推理增强上合作。RewardBench开源，易于扩展。

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引用: 23,000+ (Google Scholar)
论文: arxiv.org/abs/2511.17332

近两年代表论文

- [AAAI 2026] Agentifying Agentic AI
- [2025] Responsible Artificial Intelligence: A Comprehensive Survey
- [2025] Multi-Agent Systems for AI Governance
- [2024] Ethics and Governance in Agentic AI Systems
- [2024] Human-Centered Design for Autonomous Agents

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Agentifying Agentic AI

创新点总结

核心贡献： 本文从多智能体系统（MAS）视角重新审视当前Agentic AI的发展，提出了"Agentifying"框架。创新包括：(1)系统分析了LLM Agent与传统软件Agent的本质差异与联系；(2)提出了将Agentic AI转化为真正多智能体系统的方法论；(3)讨论了Agent化过程中的社会性、责任性、可控性挑战；(4)为构建更加可信赖、可治理的AI Agent系统提供了理论框架。Virginia Dignum是AI伦理和MAS领域知名学者，该工作代表了从责任AI视角对Agentic AI的深刻反思。

Agent	责任	MAS	WMAC
化框架	AI	视角	Invited

AAMAS Concept	Description	Agentic AI?	Role in Agentic AI
BDI Architecture	Explicit representation of beliefs, desires, intentions	❌ Not explicit	Enables interpretable, goal-driven behavior; essential for reasoning and alignment
Communication Protocols	Structured inter-agent messages with semantics	⚠ Ad-hoc, via natural language	Provides reliable coordination and prevents miscommunication in multi-agent systems
Mechanism Design / Incentive Alignment	Systems design for self-interested agents to act in desirable ways	❌ Rarely applied	Aligns agent behaviors with system-wide objectives; avoids unintended or harmful outcomes
Multi-Agent Planning / Coordination	Joint plans, task decomposition, distributed execution	⚠ Shallow coordination	Supports efficient task distribution and cooperation among agents
Negotiation / Argumentation	Agents negotiate over goals, actions, resources	❌ Absent	Enables dynamic conflict resolution and adaptive collaboration
Norms, Institutions, Roles	Agents governed by social norms or institutional rules	❌ Not modeled	Enforces behavioral constraints and fosters trust and stability in open environments
Trust / Reputation	Evaluating agent reliability over time	❌ Absent	Critical for long-term collaboration and risk-aware decision making
Social Choice / Voting	Group decision-making among agents	❌ Not used	Enables collective intelligence and democratic coordination in multi-agent scenarios
Game Theory (Co-op/Adversarial)	Strategic reasoning about others' choices	⚠ Minimal use	Improves agents' ability to operate under uncertainty and strategic interaction
Theory of Mind / Belief Modeling	Reasoning about other agents' beliefs, goals, intentions	⚠ Very rudimentary	Allows modeling and anticipation of others' behavior—key for safety, alignment, and cooperation

Agentifying框架：将Agentic AI转化为多智能体系统

合作建议

Virginia Dignum是AI伦理领域领军人物，Umeå AI Policy Lab资源丰富。适合在责任AI、AI治理、多智能体伦理上合作。WMAC特邀报告。

S

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合作机构: Gunma University, Yokohama National University
论文:
multiagents.org/2026_papers/utility_aware_task_decomposition.pdf

近两年代表论文

- [AAAI 2026] Utility-Aware Task Decomposition and Exchange across LLM Agents
- [2025] Efficient Task Allocation in Multi-Agent LLM Systems
- [2025] Utility-Based Agent Coordination Frameworks
- [2024] Task Decomposition Strategies for LLM Agents
- [2024] Multi-Agent Collaboration for Complex Problem Solving

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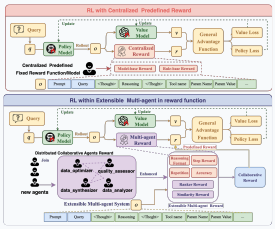
Utility-Aware Task Decomposition and Exchange across LLM Agents

创新点总结

核心贡献：本文提出了效用感知的LLM Agent任务分解与交换框架。创新包括：(1)设计了效用评估机制，量化每个子任务对不同Agent的执行价值；(2)提出了任务交换协议，允许Agent之间动态分配和交换子任务以最大化整体效用；(3)实现了去中心化的任务协调机制，避免了中心化控制带来的瓶颈问题；(4)在多领域复杂任务上验证了框架的有效性。该工作解决了多Agent协作中的任务分配效率问题，为构建高效的多Agent系统提供了新方法。

效用	任务	去中心	WMAC
感知	交换	化协调	Oral



效用感知任务分解框架：动态任务交换机制

合作建议

日本团队在多Agent协作方向有深厚积累，适合在任务分解、效用优化、去中心化协调上合作。方法实用性强，易于工程落地。

Y

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合作机构: Tsinghua University, Gaoling School
论文: arxiv.org/abs/2410.05130

近两年代表论文

- [AAAI 2026] Scalable and Accurate Graph Reasoning with LLM-Based Multi-Agents
- [2025] Multi-Agent Approaches for Graph Neural Networks
- [2025] LLM-Based Graph Reasoning: Methods and Applications
- [2024] Scalable Graph Processing with Distributed Agents
- [2024] Combining LLMs with Graph Algorithms for Reasoning

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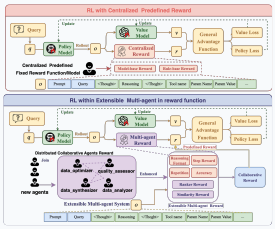
Scalable and Accurate Graph Reasoning with LLM-Based Multi-Agents

创新点总结

核心贡献：本文提出了基于LLM多智能体的可扩展图推理框架。创新包括：(1)设计了多Agent协作的图推理架构，将复杂图问题分解为可并行的子任务；(2)实现了高精度的图结构理解，LLM Agent能够准确识别图的拓扑特征；(3)提出了可扩展的推理流程，支持大规模图的高效处理；(4)在多个图推理基准上达到了SOTA性能。该工作将LLM Agent与图计算结合，为解决传统图算法的可解释性和泛化性问题提供了新思路。

可扩展	图	高精度	人大
推理	理解	SOTA	清华



多Agent图推理框架：可扩展的图结构分析方法

合作建议

人大、清华联合出品，图计算资源丰富。适合在图推理、多Agent协作、大规模数据处理上合作。arXiv已发布，代码有望开源。

Y

Yang Li

多机构合作

研究方向: 版本控制、多智能体系统、LLM工程
创新点: 将Git理念引入多Agent系统
论文: arxiv.org/abs/2511.00628
发表: AAAI 2026 WMAC Poster

近两年代表论文

- [AAAI 2026] AgentGit: A Version Control Framework for Reliable and Scalable LLM-Powered Multi-Agent Systems
- [2025] Version Control for Multi-Agent LLM Systems
- [2025] Reliable Multi-Agent Frameworks with Git-Based Coordination
- [2024] Scalable Multi-Agent Systems with Distributed Versioning
- [2024] Engineering Practices for LLM Multi-Agent Development

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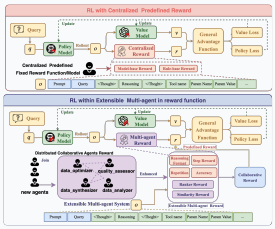
AgentGit: A Version Control Framework for Reliable and Scalable LLM-Powered Multi-Agent Systems

创新点总结

核心贡献： 本文提出了AgentGit，一个面向LLM多智能体系统的版本控制框架。创新包括：(1)将Git版本控制理念引入多Agent系统，支持Agent状态的追踪、回滚和分支管理；(2)设计了面向Agent协作的merge机制，解决多Agent并发修改冲突问题；(3)实现了可扩展的系统架构，支持大规模Agent团队的高效协作；(4)提供了可靠性和容错机制，确保系统在Agent失败时的可恢复性。该工作填补了多Agent系统工程化实践的空白，为构建可靠的多Agent系统提供了基础设施。

版本	可靠	可扩展	工程
控制	性	架构	实践



AgentGit框架：多智能体系统的版本控制机制

合作建议

AgentGit解决了多Agent系统的工程化难题，实用性强。适合在系统可靠性、大规模部署、工程实践中合作。arXiv开源，易于集成。